

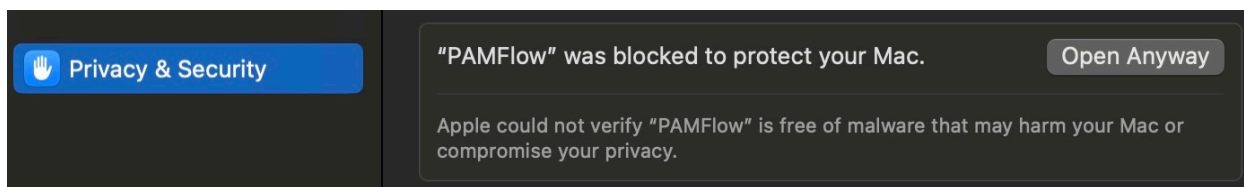
PAMFlow

User manual for audio, BRUV video, and RUV image workflows

1. Installation

MacOS may warn that the app cannot be opened because Apple cannot check it for malicious software. This is expected for test builds.

- Open the DMG and drag PAMFlow into Applications.
- If macOS blocks the first launch, open System Settings, then Privacy & Security.
- Find the PAMFlow security message and choose Open Anyway.
- Confirm Open when macOS asks again. You should only need to do this once unless you remove the app from Applications



2. Welcome And User Name

The welcome screen asks for your name. PAMFlow uses this name in export files as the Processed by or reviewer value. This makes it possible to trace who processed each sample, image, video frame, or detection event.

3. Projects

The Projects screen is your starting point after setup. Projects are grouped into In Progress and Completed, and the search box can find projects by name, recorder, status, or stored source paths.

- In Progress projects can be reopened at the next unfinished step.
- Completed projects can be inspected and exported again if their saved data is available.
- If the original project folder is missing, PAMFlow shows what is still available from saved project data and explains which resources cannot be opened.

4. New Project

Choose PAM, BRUV, or RUV, then select the raw media. You can select one folder, or select specific files.

If you select specific files, macOS may only grant PAMFlow access to those files, not permission to write beside them. PAMFlow will therefore ask for a second folder. This second dialog gives the app permission to create the project folder. You can choose the same parent folder as the raw files, or choose a different writable location such as Documents.

Project type	Recommended batch size	Notes
PAM	100 to 200 one-minute WAV samples	Manual audit can be skipped if samples are already known to be usable.
RUV	50 to 100 images	Image batches are usually quick, but large folders can still take time.
BRUV	1 to 2 ten-minute videos	SharkTrack can take around 20 minutes per video depending on video size and Mac performance.

5. Metadata

Metadata connects detections to deployment information such as opcode, date, location, depth, bottom type, and water temperature.

- Metadata can be entered manually or loaded from a CSV table.
- When a metadata CSV is uploaded, PAMFlow saves it for that project type and offers it again for future projects until a new table is uploaded.
- Use clear column names such as opcode, date, depth, bottom type, water temperature, date deployed, or date retrieved. Extra words are okay if the important field name is present.
- After selecting an opcode from an uploaded table, app fills matching metadata fields automatically.

6. Scan Overview

After project creation, PAMFlow scans the selected media and shows a quick overview. Use this screen to confirm that the expected number of files, formats, durations, sample rates, frame rates, and metadata values look right before continuing.

If a folder or drive is disconnected later, app can still show saved project records, but screenshots, audio clips, and other large resources may be unavailable until the drive is reconnected.

7. SharkTrack For BRUV And RUV

BRUV and RUV workflows use SharkTrack to find candidate animals before manual review. SharkTrack is an open-source computer-vision workflow for detecting and tracking sharks and rays in underwater imagery.

Learn more: [SharkTrack paper](#) and [website](#)

- PAMFlow shows SharkTrack progress while frames or images are processed.
- Large videos can take a while. This is a good time for a coffee, or for running a batch while the Mac is awake overnight.
- No detections is a normal outcome for many deployments. PAMFlow treats this as a completed result, not as a crash.

8. Frame Review For BRUV And RUV

Frame Review is where a person confirms whether each SharkTrack candidate is useful.

- Choose Valid when the frame contains a real animal detection you want to keep.
- Choose Invalid when the candidate is not useful. You can add a reason if helpful.
- Use Add Species to record species for valid detections.
- Use maxN to record the maximum number of individuals visible in the frame. Automatic maxN is still being improved, so check it and adjust it manually when needed.
- Finish ends the review for that project type and opens the overview.

9. PAM Manual Audit

PAM Manual Audit is a sample quality check before PAMGuard. It is not asking whether you hear a whale or dolphin. It asks whether the sample sounds like a valid underwater recording and is suitable for PAMGuard processing.

- Mark a sample Valid if it sounds usable.
- Mark Unsure or Invalid if it is noisy, corrupted, not underwater audio, or otherwise unsuitable.
- The audit is optional. You can skip it and send all samples to the PAMGuard setup step.
- If you choose Unsure or Invalid, use Add Reason to record the issue.

10. PAMGuard Setup And Detection Review

PAMFlow prepares a PAMGuard folder for acoustic processing. Open PAMGuard, load the generated template, and follow the PAMGuard help button inside PAMFlow for the exact setup steps.

- The PAMGuard help button explains database selection, binary storage, and choosing the input audio folder.
- After PAMGuard finishes, return to PAMFlow and import the PAMGuard results.
- PAMGuard may produce many raw click records. PAMFlow groups them into reviewable detection events so the reviewer is not asked to review every individual click.
- Clicks close together are grouped into click bouts. Very short or sparse bouts are filtered out.
- Click trains and whistles are prioritised above click bouts. Individual clicks are used as evidence, not standalone review items.
- Events that overlap, or are within a short gap in the same recording, are merged so duplicate windows are avoided.
- Each event gets a review window around the evidence, then the reviewer marks it Valid, Unsure, or Invalid.

11. Completion And Export

The completion screen summarises what was reviewed and lets you choose which fields appear in the exported CSV. Use Customise Fields to add, remove, or reorder columns. PAMFlow remembers export field choices per project type.

- For BRUV and RUV, screenshots are stored in the project folder under detections. BRUV creates a folder per video; RUV stores screenshots directly in detections.
- Reports are saved in the project folder.
- For PAM, Export Detections creates samples.csv, detection_events.csv, a samples folder, high-resolution spectrogram images, original WAV samples when available, and one Raven selection table per sample.
- The Raven selection table is per sample, not per event, so every event in a sample can be highlighted together in Raven.
- If resources are missing, PAMFlow exports the saved data it still has and communicates what could not be included.

12. Settings And Feedback

Settings contains appearance controls and feedback. PAMFlow defaults to dark mode, but you can switch to light mode or system mode.

Use Send Feedback to report bugs, issues, and suggestions. You can include screenshots or small attachments so the developer can understand what happened.

13. Planned Features

- Improved individual tracking in BRUV video using SharkTrack live tracking settings, so repeated frames can be connected to the same animal more reliably.
- MaxN calculations for each frame
- Saving and resuming long processing progress if the app or Mac is interrupted.
- Google Sheets integration for metadata lookup and export delivery
- More project types and additional export presets.